MIL-P-36668B
31 March 1977
SUPERSEDING
MIL-P-0036668A (DSA-DM)
20 December 1973 and
MIL-P-36668 (DSA-DM)
17 March 1970

MILITARY SPECIFICATION

PIPET, SEROLOGICAL, STERILE, DISPOSABLE, GLASS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a disposable, plugged, 1 milliliter capacity sterile glass serological pipet, suitable for use during laboratory procedures.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing).
MMM-A-250	-	Adhesive, Water-Resistant (for Closure of Fiberboard Boxes).
DD-G-541	-	Glass (Laboratory).
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Fiberboard.
PPP-B-676	-	Boxes, Setup.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, CODE ATT, 2800 S. 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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MILITARY

MIL-S-36586 - Sterilization Test Strip Set, Bacterial Spore.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Standard 602 - Testing of Glass Volumetric Apparatus.

(Application for copies should be addressed to Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Glass shall be borosilicate in accordance with DD-G-541, Type 1, Class A or B.

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3.2 Construction. Pipet shall be of one-piece glass (see 3.1) construction. Any cross section of the pipet bore taken in a plane perpendicular to the longitudinal axis shall be circular. Pipet shall have a mouth section at one end and a tapered delivery tip at the other end. The delivery tip shall have no constriction of the bore at the orifice. At the mouth end of the pipet, there shall be a plug formed from suitable cotton or synthetic material, in contact with the surface of the bore for a length of at least 15 millimeters (mm) so as to afford adequate protection to the user.

3.3 Scale. Pipet shall have a capacity of 1 milliliter (ml) and shall be marked with a scale graduated from 0 to 1, with 0 at the top (mouth end) of the scale. The area between 0 and 0.9 shall have graduation markings as follows: least graduations hundredths, intermediate graduations five hundredths, and main graduations tenths. The segment between 0.9 and the delivery tip need not be marked with graduation lines or numbers.

3.3.1 Graduation markings. Graduation markings shall be sharply defined continuous lines in a plane perpendicular to the longitudinal axis of the pipet and parallel to each other. The length of the graduation lines shall increase in the following order: least, intermediate and main. Graduation lines shall extend around the pipet circumference for the following minimum distances: least 1/7, intermediate 1/5, and main 3/5. Main graduations ("0" and 0.1 ml increments) shall be marked with Arabic numerals directly above the line referenced.

3.4 Dimensions. Dimensions, in millimeters, shall be as follows:

Overall length	273.0 ± 20.0
Length of scale (0 to 1)	140.0 minimum
Outside diameter	4.6 ± 0.5
Wall thickness (mouth end of pipet)	0.9 minimum
Length from mouth end to top of scale	90.0 minimum

3.5 Marking.

3.5.1 Identification marking. Each pipet shall be permanently marked (see 3.8) with the manufacturer's name or registered trademark, capacity and method of use (to deliver). "To deliver" may be abbreviated "T.D.".

3.5.2 Last drop blow-out marking. A single, circular wide band or more than one thin band shall be permanently marked at the mouth end section of the pipet to indicate that the last drop is to be blown out (See 3.8).

3.6 Accuracy. Pipet shall have an accuracy of ± 0.02 ml, when tested as specified in 4.4.1.

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3.7 Outflow time. Outflow time shall be between 0.5 and 2 seconds when tested as specified in 4.4.2.

3.8 Permanency of markings. Markings shall be legible and distinguishable after tests specified in 4.4.3.

3.9 Sterility. Pipet shall be sterile when tested with a biological indicator as specified in 4.4.4.

3.9.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, or inoculated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Phila., PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.9.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.10 Workmanship. Workmanship shall be first class throughout. Pipet shall be free of defects which detract from its appearance or may impair its serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the glass, may be furnished in lieu of actual performance f such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with table I and MIL-STD-105. Unit of product shall be the pipet.

TABLE I. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	S-4	2.5
Minor defects	S-4	4.0
For dimensional examination	S-2	4.0

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with table II and MIL-STD-105. Unit of product shall be the pipet, except the unit of product for sterilization testing which shall be that quantity of pipets that will be supplied in the intermediate package (see 5.1.2). Acceptance number for all testing shall be zero.

TABLE II Sampling for tests.

Characteristic	Requirement	Procedure	Inspection level
Accuracy	3.6	4.4.1	S-2
Outflow time	3.7	4.4.2	S-2
Resistance to water	3.8	4.4.3.1	S-1
Resistance to alcohol	3.8	4.4.3.2	S-1
Sterility	3.9	4.4.4	S-3

4.3 Examination. The pipet shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following table:

TABLE III Classification of defects.

Category	Defect*
Major:	
101	Pipet not free of sharp edges.
102	Pipet not free of chips.
103	Pipet not free of cracks or breakage.
104	Pipet not free of protrusions.
105	Marking on pipet missing, incomplete, incorrect, or illegible (except for manufacturer's identification).
106	Cotton or synthetic plug missing.
107	Graduations not in a plane perpendicular to the longitudinal axis and parallel to each other.
108	Delivery tip not free of constriction of bore at orifice.
Minor:	
201	Pipet not free of bubbles.
202	Manufacturer's identification marking missing, incomplete, incorrect, or illegible.
203	Graduation markings not sharply defined continuous lines.

* Examination shall not be restricted to the defects listed herein.

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4.3.2 Dimensional examination. The pipet shall be examined for defects in dimension. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy. Accuracy of graduation markings shall be tested in accordance with methods outlined in NBS Circular 602. No less than two graduate markings shall be tested, one at full capacity and the other at a level no greater than half capacity.

4.4.2 Outflow time. Outflow time shall be tested in accordance with methods outlined in NBS Circular 602.

4.4.3 Permanency of markings.

4.4.3.1 Water. Pipet shall be submerged in distilled water for 15 minutes at 25° Centigrade (C) $\pm$ 5°.

4.4.3.2 Alcohol. Pipet shall be submerged in 95% ethyl alcohol for 15 minutes at 25° C $\pm$ 5°.

4.4.4 Sterility test. Pipet shall be tested for sterility in accordance with the method specified in the USP on Procedured Details for Sterility Tests for Non-Parenteral Devices, using inoculated product or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product or paper spore strip (carrier). The incubation temperature shall be as indicated in 4.4.4.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in the table of 4.2.2.

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4.4.4.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65° C
Dry heat	B. stearothermophilus or	55° - 65° C
	B. subtilis var. niger (or globigii)	30° - 35° C
Gas	B. subtilis var. niger (or globigii)	30° - 35° C
Radiation	B. pumilus	30° - 35° C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.4.2 Inoculated carrier. Inoculated carriers (paper spore strips), if used, shall be of equivalent resistance to spore populations included in MIL-S-36586. The method of incubation and number of test units shall be as indicated in 4.4.4 and 4.4.4.1. The method of sterilization will control the organism in the spore strip to be tested for, as shown in 4.4.4.1.
NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by USP.

4.4.4.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the contracting officer, and approval obtained before employment of the alternate method on a Government contract.

4.4.4.4 Product testing. Testing for sterility of the product after receipt by the Government shall be without a biological indicator and in accordance with the USP.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Unit of issue. One package (25 pipets), as specified, constitutes one unit of issue.

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5.1.2 Level A.

5.1.2.1 Unit package. Each pipet shall be packaged in a sealed waterproof tear-type container which shall maintain sterility of the contents unless the container is damaged or opened. Twenty-five individually packaged pipets shall be overpacked in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676 except that commercial colors are acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.2.2 Intermediate package. Ten unit packages shall be packaged in a box constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2-inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.3 Level C. Unit shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Five intermediate packages (50 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B. Five intermediate packages (50 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Sterility lot (control) number, contract or purchase order number, name of contractor shall be shown. The method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry heat sterilization) shall be shown following the level of packaging and date of packaging. Example: "A-9/72-G." Final sterilization shall be on or after indicated packaging date. In addition, each individual container or polyethylene bag containing 25 pipets shall be marked with sterility lot number, and the legend: "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED," and with the requirements of the Federal Food, Drug and Cosmetic Act.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. Sterility lot (control) number, contract or purchase order number and name of contractor shall be shown. In addition, the method of sterilization as specified in 5.3.1 shall also be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Sterility lot (control) number and the method of sterilization as specified in 5.3.1 shall be shown.

6. NOTES

6.1 Intended use. Pipet shall be suitable for use in serological laboratory procedures.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.4 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-926-7713	PIPET, SEROLOGICAL, Disposable, Sterile, 1 ml, 25s

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6.5 Limitations. This specification does not cover all types of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project No. 6640-1010

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-P-36668B PIPET, SEROLOGICAL, STERILE, DISPOSABLE, GLASS

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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